

Measuring and Removing an Object’s Light Footprint

Reflection-Aware Object Removal in 3D Gaussian Splatting via a Render–Edit–Refit Pipeline

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1 Problem Definition

3D Gaussian Splatting (3DGS) [1] is widely used both for scene reconstruction and for downstream tasks such as 3D object editing. I want to remove an object from a reconstructed 3DGS scene so that the result still looks correct from every viewpoint. This is hard because removing an object does not only mean filling the hole where it stood: the object also affects the rest of the scene — it casts a shadow, it can be reflected in a nearby glossy or mirror surface, and it can bounce coloured light onto walls and other surfaces. If I simply delete the object’s Gaussians, these effects stay baked into the surrounding surfaces, and the scene becomes physically impossible: the vase is gone, but its shadow and its reflection in the table are still there.

I model the colour of a background point as

$$L = L_{\text{direct}} + L_{\text{obj}}, \tag{1}$$

where L_{obj} is the part of the light caused by the object (blocked light gives a shadow, reflected light gives a mirror image, bounced light gives colour bleed). Ordinary removal keeps L as a fixed texture, so L_{obj} survives. My goal is to remove the object *and* recover the clean background L_{direct} , consistently across views.

2 Related Work

Object removal / 3D inpainting. Several methods remove objects from radiance fields and inpaint the hole in a view-consistent way: SPIn-NeRF [2], Weder et al. [3] and OR-NeRF [4] for NeRF, and GScream [5] and Gaussian Grouping [6] for 3DGS. They usually lift 2D masks (e.g. from SAM [7]) into 3D, delete that region, and fill it using a 2D inpainter with depth for consistency. Their limit is that they treat the surrounding scene as fixed appearance: they fix geometry and texture but do not update lighting effects, so the object’s shadow and reflection are left behind.

Footprint-aware removal. A few recent works start to address the leftover effects. GauComp [8] segments and removes an object’s cast shadow together with the object, and GOR-IS [9] removes objects in an intrinsic (material/lighting) space to keep global illumination consistent

— but both focus mainly on shadows, and their validation is still primarily perceptual. Full inverse rendering (e.g. IRGS [15]) recomputes light transport physically, but it is heavy and built for relighting rather than removal. A recent study also shows that 2D image editing models have a limited understanding of lighting effects [20], which suggests per-frame image priors alone are not enough for this task.

Measuring what removal leaves behind. SPIn-NeRF [2] introduced paired scenes captured with and without the target object, and Remove360 [13, 14] shows with real 360° captures that current removal methods leave residuals that foundation models can still detect. However, Remove360 measures *semantic* residuals (can a model still infer what was removed?), not the physical light footprint, and to my knowledge no benchmark isolates the residual into its physical components (shadow vs. reflection vs. colour bleed) with exact ground truth. My Blender-based validation setup (Section 5) provides exactly this controlled setting.

Diffusion-based editing of 3D scenes. Instruct-NeRF2NeRF [10] and its 3DGS version Instruct-GS2GS [11] edit a reconstructed 3D scene with a text instruction: a 2D diffusion editor (InstructPix2Pix [16]) repeatedly edits the training views while the 3D model keeps being optimised on them (“iterative dataset update”), so the 3D representation itself enforces one consistent edit. This shows that a 2D generative prior can drive a 3D edit without any physical lighting model.

Video diffusion models for removal. The closest work to my goal is ROSE [18], a video inpainting model explicitly trained to remove objects *together with* five side effects (shadows, reflections, light, translucency, mirror images), using paired synthetic videos generated with a 3D rendering engine. General-purpose video editors such as VACE [12] also perform instruction- and mask-conditioned removal with strong temporal consistency learned from real videos. Independently, Chang et al. [17] show that much of the frame-to-frame inconsistency of diffusion editing comes from sampling *independent* noise per frame, and that warping a single noise field along the motion between frames (*f*-noise) makes the outputs temporally consistent. All of these, however, operate purely on 2D video: they produce edited frames, not an editable, view-consistent 3D scene.

Gap. 3D removal methods handle the hole but not the light footprint; video models can now remove an object with its footprint, but only in 2D; and no controlled benchmark measures the footprint per effect in 3D. My project connects these threads: use a video diffusion prior to clean up the object’s light footprint, use 3DGS to make the result truly 3D-consistent, and measure the result with exact synthetic ground truth.

3 Approach

Instead of modelling the lighting physically, which may be intractable in my setting, I outsource the cleanup to a learned video prior, and use 3DGS twice — once before editing and once after. I first render the reconstructed 3DGS scene into a video by moving a virtual camera around the scene; a pre-trained video model then edits this video to remove the object; finally I re-fit a fresh 3DGS on the edited frames. The approach is possible because I generate the camera path myself, so every frame has an exactly known pose (and depth), with no need for COLMAP.

The pipeline has four steps:

1. **Reconstruct.** Input: the scene images. I fit a standard 3DGS model. Output: a 3D model of the scene with the object still present.

2. **Render an orbit.** I render the fitted 3DGS along a smooth camera path. Output: a coherent “video” of the scene, plus the exact camera pose and depth map of every frame (used later for noise warping).
3. **Edit the video.** Input: the orbit video, a mask, and a removal instruction. A pre-trained video editing/inpainting model removes the object; the hypothesis is that its learned prior removes the object *together with* its shadow, reflection, and colour bleed. The object mask comes from SAM [7] on one frame, propagated across frames with SAM 2 [19]. Because the reflection lies outside the object mask, I will either (a) let the editor modify a full-frame region and protect untouched areas with a keep-loss at re-fit time, or (b) annotate the reflection/shadow region once and propagate it with SAM 2; I will try (a) first. As the editing model I will use ROSE [18], which is purpose-built for removing side effects, with VACE [12] as the general-purpose alternative (also used in the ablations), subject to released weights fitting a Colab GPU.
4. **Re-fit.** Input: the edited frames with their known poses. I train a fresh 3DGS as if they were captured photos, with the standard photometric loss on edited frames plus a keep-loss on unedited regions against the original renders, so the untouched part of the scene cannot drift. Output: a clean 3D scene renderable from any viewpoint; small frame-to-frame disagreements are averaged out by the 3D representation.

Why this could work. Plain removal with 2D inpainting fixes geometry and texture but leaves L_{obj} baked in, because the inpainter only sees the hole. A video model sees the whole frame sequence and has learned from data what scenes look like *without* the object — in ROSE’s case, it was explicitly trained for this — so shadow and reflection removal come from the prior rather than from explicit ray tracing. Everything is sized for a single Google Colab GPU: 480p resolution, orbits of ~ 81 frames, one compact scene with one object and at least one glossy surface.

Extension 1: In-context examples from Blender

The editor may remove the object but leave its footprint, since the instruction “remove the vase” does not explicitly say “...and its reflection.” To demonstrate what a *complete* removal looks like, I will condition the model with an in-context example: a pair of videos of the same synthetic scene rendered in Blender *with* and *without* an object, including a glossy surface so the pair shows reflections and shadows disappearing too. The pair is prepended to the model’s context as a before/after demonstration. The Blender infrastructure I need for validation produces these pairs for free. This use of reference conditioning is off-label for current video editors, so it is an explicitly experimental extension whose result I will report either way.

Extension 2: Multi-view consistency

Step 3 is the weak link: diffusion editing of different frames can disagree (flicker, different fill content), and a 3DGS trained on inconsistent images becomes blurry or ghosted. I will explore two remedies:

- **Iterative dataset update (Instruct-GS2GS style [11]).** Rather than editing the video once, alternate between editing individual rendered views and re-optimising the Gaussians

on them. The 3D model acts as the consistency mechanism, pulling all edits toward one coherent scene.

- **Warped noise (f -noise [17]).** Make the diffusion *noise* consistent instead of correcting the outputs: sample one noise field and warp it from frame to frame following the camera-induced motion, which I can compute exactly from my known poses and rendered depth (no optical flow needed). If the same 3D point receives the same noise in every view, the edits agree across views by construction.

4 Motivation

Realistic object removal (“diminished reality”) is useful for AR/VR editing, film clean plates, removing people from scans for privacy, and editing scenes for simulation. In all of these, a leftover shadow or reflection of an object that is no longer there is exactly what the eye notices first. As 3DGS becomes a common scene format, making its edits physically plausible (not just geometrically complete) is a small but real missing piece: recent benchmarks confirm that residuals are a measurable problem [13, 14], and the appearance of ROSE [18] shows the community considers footprint removal important — but only in 2D video. Compared to explicit inverse rendering, my pipeline replaces physical lighting computation with a learned prior, which is feasible on a single GPU and reuses strong pre-trained models.

5 Validation

I will mainly use synthetic scenes rendered in Blender, which serve double duty: rendering the same camera orbit *with* and *without* the object gives an exact ground-truth clean plate for evaluation, and the same infrastructure produces the in-context pairs for Extension 1. Differencing the two renders (outside the object’s silhouette, via the renderer’s object index pass) automatically yields per-effect footprint masks — the shadow region, the reflection region, and the colour-bleed region — with no manual annotation. Everything is sized for one Colab GPU: one compact scene, moderate resolution, at least one glossy or mirror surface, one object to remove. One real capture will be added as a qualitative demo if time allows.

Mid-point checks.

- Run the basic pipeline end to end once (3DGS → orbit render → video edit → 3DGS re-fit) and verify the loop closes, even if quality is imperfect.
- *Measure the problem:* show that plain deletion + 2D inpainting leaves visible reflection/shadow residue against the clean plate, establishing the baseline gap.

Final checks.

- **Quantitative:** PSNR / SSIM / LPIPS against the clean plate — full image and masked to each footprint region (shadow / reflection / bleed) separately, in the spirit of Remove360 [13]. Baseline: a GScream-style [5] removal+inpaint result (published numbers may be cited rather than re-run, as allowed by the project rules). Evaluated both on the orbit and on held-out novel viewpoints rendered from the re-fitted 3DGS, so 3D consistency is part of the score.

- **Ablations:** (a) editing model: ROSE vs. VACE vs. per-frame image editing; (b) with vs. without in-context Blender examples; (c) consistency mechanism: none vs. iterative dataset update vs. warped noise, measured by re-fit sharpness and warping error between edited frames.
- **Qualitative:** before/after orbit videos showing the object, its reflection and its colour bleed disappearing together.

As the project guidelines note, a clearly evaluated negative result (e.g. if the video prior consistently leaves specular residue, or if noise warping does not transfer to view changes as well as to temporal motion) is a fine outcome, and I will report it honestly.

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